Big Data Management® 10.2.2: Security for Amazon EMR

In this article, learn the security functionalities that Big Data Management supports and how to configure them in a Amazon EMR environment.

Overview

Informatica Big Data Management provides a holistic support for the security layers of the Hadoop cluster in order to keep your data and your enterprise secure. To integate Big Data Management with a secure Hadoop cluster, it is not mandatory to configure security on the Informatica domain.

This article discusses Big Data Management support for authentication, authorization, and encryption mechanisms that an Amazon EMR cluster uses.

The following list describes types of encryption that you can apply to data in the Amazon EMR environment:

- In-transit **key pair encryption**: Use a key pair for encryption, either creating a key pair in AWS or import your own public key. See AWS documentation.
- In transit **custom encryption**: Use a TLS custom certificate provider to encrypt in-transit data. See AWS documentation.
- At-rest **CSE-KMS** encryption: Use a customer master key (CMK) stored in AWS Key Management Service (AWS KMS). See AWS documentation.
- At-rest **CSE-custom** encryption: Use a master key you store within your application. See AWS documentation.
  Note: This requires you to manually copy the .jar file to the Data Integration Service machine. See the Data Engineering Integration documentation.
- At-rest **SSE-S3** encryption: When you use server-side encryption (SSE) of S3 data, you request Amazon S3 to encrypt data, using a master key, before writing it to S3 buckets and decrypting it when you read from S3. See AWS documentation.
- At-rest **SSE-KMS** encryption: Use customer master keys (CMK) for authorization when encrypting and decrypting data in S3 buckets. See AWS documentation.
- At-rest **encryption of local disks using KMS**: Encryption of data stored on local disks using a KMS customer-managed key (CMK). See AWS documentation.
- Specify a custom AMI to encrypt the EBS root device volumes of EC2 instances. See AWS documentation.

Documentation Reference

For more information, refer to the following documentation:

- Data Engineering Integration Guide
- S3 Adapter Guide
- Redshift Adapter Guide
- Snowflake Adapter Guide
- HBase Adapter Guide

Supported Security Systems for Amazon EMR Clusters

The Spark and Blaze engines have identical supported security systems. The following table lists the supported security systems for the run-time engines in the Hadoop environment:

Encryption Type	Encryption	Authentication	Authorization
In-transit encryption of data on the AWS network	• Key pair encryption using .pem file • Custom encryption using a TLS custom certificate	Kerberos authentication (Cluster-Dedicated KDC)	EMRFS Role-based authorization enabled
At-rest encryption of data in S3	• CSE-KMS • CSE-custom • SSE-S3 • SSE-KMS		
At-rest encryption of data on local disks	At-rest encryption of local disks using KMS		

[1] Kerberos support with MIT-type Kerberos, no AD/LDAP server.

Supported Informatica Adapters for Amazon EMR Clusters

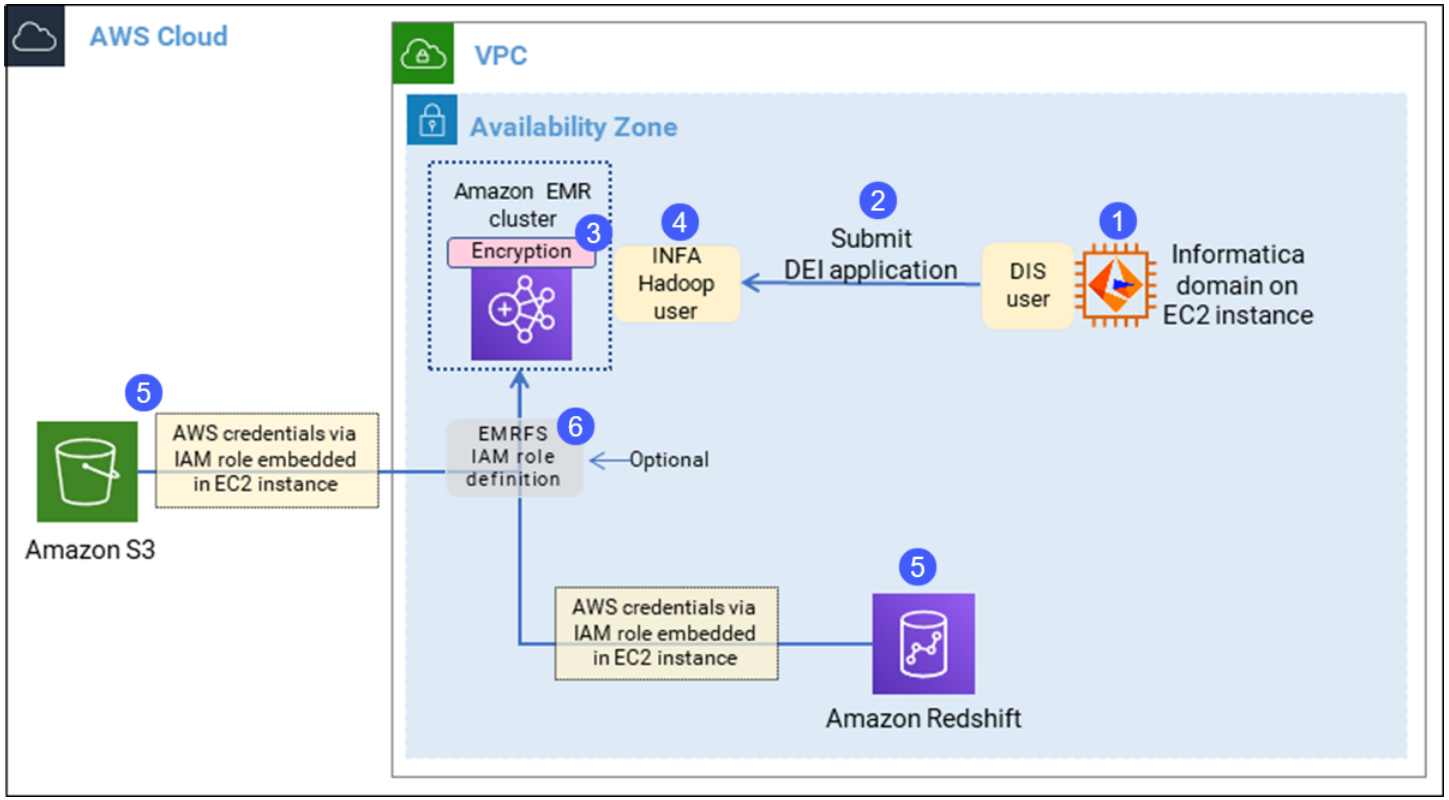
The following table shows support by adapters for various security configurations:

Adapter	In-Transit Encryption	At-Rest Encryption
S3 Redshift	Key pair (.pem)	CSE-KMS; SSE-S3; SSE-KMS; encryption for local disks
Snowflake HBase	Key pair (.pem)	CSE-KMS; CSE-custom; SSE-S3; SSE-KMS; encryption for local disks
n/a	EBS root volume encryption	

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Security Configuration for Amazon EMR

The following flowchart shows security configuration options for a deployment of Data Engineering Integration on AWS integrated with an Amazon EMR cluster.



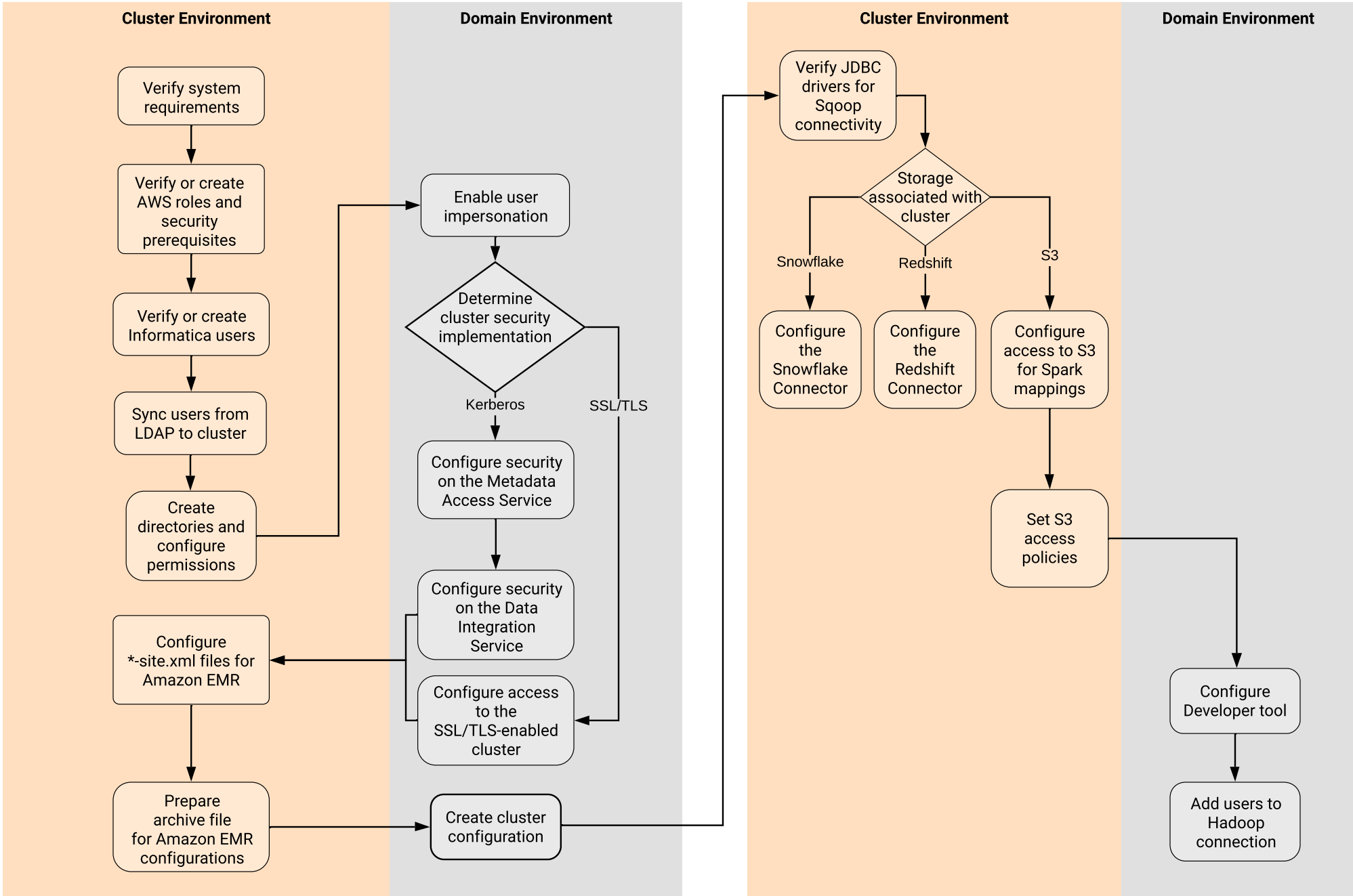
The following list of elements matches the numbered elements in the image:

1. Informatica domain deployed on an EC2 instance in an AWS VPC.
2. The Data Integration Service submits a job to the cluster as the Data Integration Service user.
3. The Amazon EMR cluster is secured with Kerberos (MIT) and one of several encryption options (see page 1).
4. The EMR cluster executes jobs as the Informatica Hadoop user.
5. The EMR cluster accesses resources such as S3 and Redshift using credentials certified by IAM roles set up for the cluster in the AWS console. AWS embeds these roles in each EC2 instance where resources reside.
6. Optionally enable EMRFS role-based authorization to define access to storage resources.

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Security Configuration for Amazon EMR

The following flowchart shows the Big Data Management security configuration tasks and where to perform the tasks. Click on the task to see the corresponding topic in documentation.



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